MAE 6246 - Linear Systems and Control

Week 10: State Observers and Output Feedback

Fall 2026

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Lecture: Friday, October 30, 2026

Main question: How can we implement state feedback when only some combinations of the state are measured?

Scope: model prediction and measurement correction; observer error dynamics; pole placement through duality; observable canonical coordinates; detectability; observer-based feedback and the separation principle; noise sensitivity and simulation.

Prerequisites: observability and the PBH test; controllability decomposition; state-feedback pole placement; similarity transformations.

1 Why estimate the state?

1.1 From state feedback to measured outputs

Last week we designed $u = -Kx$ to make $A - BK$ Hurwitz. In practice, the controller may receive only the output

$$\dot{x} = Ax + Bu, \qquad y = Cx, \quad x \in \mathbb{R}^n,\ u \in \mathbb{R}^m,\ y \in \mathbb{R}^p.$$

For example, a position sensor on a moving mass measures position but does not directly measure velocity. We seek an estimate $\hat{x}$ using the known input u and measured output y, then implement $u = -K\hat{x}$.

An observer has its own evolving state $\hat{x}$. It reconstructs information through the model and output history; it does not require algebraically inverting C. In particular, $p < n$ does not by itself prevent observability.

1.2 What happens if we only simulate the model?

A model predictor uses

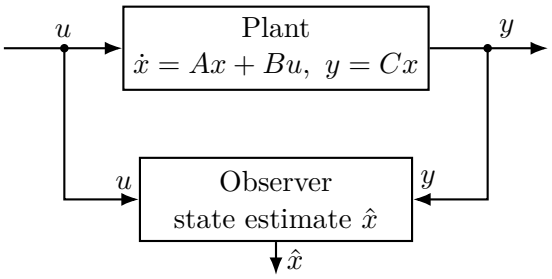
$$\dot{\hat{x}} = A\hat{x} + Bu.$$

With an exact model, the same input, and $\hat{x}(0) = x(0)$, it reproduces the true state exactly. However, the initial state is usually not known. Defining

$$\boxed{e = x - \hat{x}},$$

we obtain $\dot{e} = Ae$ and $e(t) = e^{At}e(0)$. The input cancels, so we cannot choose the error dynamics independently. If A is Hurwitz, initial error decays naturally; if A has undamped or unstable modes, it need not decay. Model errors and disturbances create further discrepancies.

1.3 Use both the input and the measured output



The observer receives the input actually applied to the plant. Its estimate is corrected by comparing the measured output with the output predicted from $\hat{x}$.

2 The Luenberger observer and its error dynamics

2.1 Prediction plus measurement correction

The predicted output is $\hat{y} = C\hat{x}$. The residual $y - C\hat{x}$ measures disagreement between prediction and measurement. The full-order Luenberger observer is

$$\boxed{\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}), \qquad L \in \mathbb{R}^{n\times p}.}$$

The first two terms predict motion using the model. The gain L converts the p output residuals into corrections to the n estimated-state derivatives. It is a design matrix, not an inverse of C.

We initially assume an exact model, a known applied input, and noiseless measurements. If the plant has known direct feedthrough $y = Cx + Du$, use $y - C\hat{x} - Du$ as the residual; the error calculation below is unchanged.

2.2 Deriving autonomous error dynamics

Using $e = x - \hat{x}$ and $y = Cx$,

$$\begin{aligned}\dot{e} &= (Ax + Bu) - \big(A\hat{x} + Bu + L(Cx - C\hat{x})\big)\\ &= (A - LC)(x - \hat{x}).\end{aligned}$$

Thus

$$\boxed{\dot{e} = (A - LC)e, \qquad e(t) = e^{(A-LC)t}e(0).}$$

The estimate converges to the true state for every initial error if and only if $A - LC$ is Hurwitz. This is an asymptotic statement: a continuous-time observer generally does not recover an unknown initial state exactly after a finite waiting time.

2.3 Observer design is a dual pole-placement problem

Recall that (A, C) is observable if and only if (A^T, C^T) is controllable. Since

$$(A - LC)^T = A^T - C^T L^T,$$

we can apply state-feedback pole placement to the dual system. If a dual gain K_d gives the desired eigenvalues of $A^T - C^T K_d$, choose

$$\boxed{L = K_d^T.}$$

Observer pole-placement theorem. All eigenvalues of $A - LC$ can be assigned arbitrarily if and only if (A, C) is observable. Desired nonreal poles must occur in conjugate pairs for a real gain L.

This theorem applies to any number of outputs. The single-output canonical construction on the next page gives a direct coefficient-matching proof.

3 Pole placement in observable canonical coordinates

3.1 A construction for one measured output

Assume $p = 1$ and (A, C) is observable. Retain the coordinate convention $x = Qz$ from Week 9. We can choose Q so that

$$\bar{A} = Q^{-1}AQ = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{bmatrix}, \qquad \bar{C} = CQ = \begin{bmatrix} 0 & \cdots & 0 & 1 \end{bmatrix},$$

where $\det(sI - A) = s^n + a_{n-1}s^{n-1} + \cdots + a_0$. Transforming the estimate as $\hat{x} = Q\hat{z}$ gives $\bar{L} = Q^{-1}L$.

For the desired polynomial

$$p_o(s) = \prod_{j=1}^{n}(s - \nu_j) = s^n + d_{n-1}s^{n-1} + \cdots + d_0,$$

choose

$$\boxed{\bar{L} = \begin{bmatrix} d_0 - a_0 \\ d_1 - a_1 \\ \vdots \\ d_{n-1} - a_{n-1} \end{bmatrix}, \qquad L = Q\bar{L}.}$$

The product $\bar{L}\bar{C}$ changes only the final column of $\bar{A}$, replacing $-a_j$ by $-d_j$. Therefore $\bar{A} - \bar{L}\bar{C}$ has characteristic polynomial p_o.

3.2 Constructing and checking the coordinate transformation

Form the observability matrices

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}, \qquad \bar{\mathcal{O}} = \begin{bmatrix} \bar{C} \\ \bar{C}\bar{A} \\ \vdots \\ \bar{C}\bar{A}^{n-1} \end{bmatrix}.$$

Because $\bar{C}\bar{A}^j = CA^jQ$, we have $\bar{\mathcal{O}} = \mathcal{O}Q$, hence

$$\boxed{Q = \mathcal{O}^{-1}\bar{\mathcal{O}}.}$$

Check $AQ = Q\bar{A}$ and $CQ = \bar{C}$. Numerically, solve $\mathcal{O}Q = \bar{\mathcal{O}}$ rather than explicitly forming an inverse.

3.3 Distinguish single-output formulas from the general theorem

For $p > 1$, $\mathcal{O} \in \mathbb{R}^{np \times n}$ is rectangular; the inverse above is not defined. The observer pole-placement theorem still applies through the dual pair (A^T, C^T). Use a multi-input pole-placement algorithm on that pair. An arbitrary canonical target is not made valid by replacing an inverse with a pseudoinverse.

4 Worked example: estimating position and velocity

4.1 The model and observability test

Consider the same second-order model used for feedback design:

$$A = \begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

Here x_1 is position, x_2 is velocity, and $y = x_1$. Since

$$\mathcal{O} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

the system is observable from position alone. The dynamics relate the time variation of position to velocity; there is no need to differentiate a noisy position measurement directly.

4.2 Choose observer poles and match coefficients

Let $L = [\ell_1, \ell_2]^T$. Then

$$A - LC = \begin{bmatrix} -\ell_1 & 1 \\ -2 - \ell_2 & 0 \end{bmatrix}, \qquad \det(sI - A + LC) = s^2 + \ell_1 s + (2 + \ell_2).$$

For observer poles $-5, -10$,

$$p_o(s) = (s + 5)(s + 10) = s^2 + 15s + 50.$$

Matching coefficients yields

$$\boxed{L = \begin{bmatrix} 15 \\ 48 \end{bmatrix}, \qquad A - LC = \begin{bmatrix} -15 & 1 \\ -50 & 0 \end{bmatrix}.}$$

The implementable observer is

$$\boxed{\begin{aligned} \dot{\hat{x}}_1 &= \hat{x}_2 + 15(y - \hat{x}_1), \\ \dot{\hat{x}}_2 &= -2\hat{x}_1 + u + 48(y - \hat{x}_1). \end{aligned}}$$

Every quantity on the right is either known, measured, or an observer state.

4.3 Check using canonical coordinates

Choose $Q = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Then

$$\bar{A} = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}, \quad \bar{C} = [0, 1], \quad \bar{L} = \begin{bmatrix} 50 - 2 \\ 15 - 0 \end{bmatrix} = \begin{bmatrix} 48 \\ 15 \end{bmatrix},$$

and $L = Q\bar{L} = [15, 48]^T$, agreeing with direct matching. The entries of L change with the state-coordinate order; the observer poles do not.

5 Unobservable modes and detectability

5.1 Which estimation errors can measurements correct?

An unobservable initial-state difference produces no output difference, so measurement correction cannot distinguish it. For a nonzero vector v satisfying

$$Av = \lambda v, \qquad Cv = 0,$$

we have $(A - LC)v = Av = \lambda v$ for every L. The corresponding eigenvalue cannot be moved by an observer gain.

The full statement follows from an observability decomposition. Choose $x = Qz$ with the final basis vectors spanning the unobservable subspace. Then

$$\bar{A} = \begin{bmatrix} A_o & 0 \\ A_{21} & A_u \end{bmatrix}, \qquad \bar{C} = [C_o \; 0], \qquad \bar{L} = \begin{bmatrix} L_o \\ L_u \end{bmatrix},$$

where (A_o, C_o) is observable. Therefore

$$\bar{A} - \bar{L}\bar{C} = \begin{bmatrix} A_o - L_o C_o & 0 \\ A_{21} - L_u C_o & A_u \end{bmatrix}.$$

The unobservable-block eigenvalues remain fixed, just as uncontrollable eigenvalues remain fixed under state feedback.

5.2 Detectability is enough for a convergent observer

The pair (A, C) is *detectable* if some L makes $A - LC$ Hurwitz. The decomposition gives

$$\boxed{(A, C) \text{ detectable } \iff A_u \text{ is Hurwitz.}}$$

Equivalently, the PBH matrix must have full column rank for every nondecaying mode:

$$\boxed{\operatorname{rank} \begin{bmatrix} \lambda I - A \\ C \end{bmatrix} = n \quad \text{for all } \lambda \in \mathbb{C} \text{ with } \operatorname{Re}\lambda \geq 0.}$$

Only eigenvalues of A in that region need checking. Detectability is dual to stabilizability. Observability permits arbitrary pole assignment; detectability only guarantees that a stable error dynamics can be obtained.

5.3 Example: an unseen state that decays

Let $A = \operatorname{diag}(1, -2)$ and $C = [1, 0]$. The second state is unobservable, but its estimation error decays naturally. Taking $L = [4, 0]^T$ gives $A - LC = \operatorname{diag}(-3, -2)$. If the unseen eigenvalue were 0 or $+2$, no choice of L could make all initial errors converge to zero.

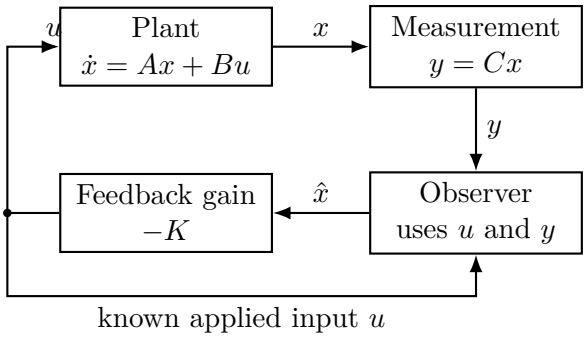
Decay of unseen errors does not imply that the initial state was uniquely reconstructed. It only means that its remaining influence disappears asymptotically.

6 Observer-based feedback and the separation principle

6.1 Closing the loop using the estimated state

Replace unavailable state feedback by

$$\boxed{u = -K\hat{x}, \qquad \dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}).}$$



This is a dynamic output-feedback controller: it uses y and maintains the internal state $\hat{x}$. The observer is driven by the same input used by the plant.

6.2 Use the estimation error as a coordinate

Because $\hat{x} = x - e$,

$$\dot{x} = Ax - BK(x - e) = (A - BK)x + BKe.$$

The error equation remains $\dot{e} = (A - LC)e$. Consequently,

$$\boxed{\begin{bmatrix} \dot{x} \\ \dot{e} \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x \\ e \end{bmatrix}.}$$

The block triangular structure proves

$$\boxed{\operatorname{spec}(A_{\text{combined}}) = \operatorname{spec}(A - BK) \cup \operatorname{spec}(A - LC),}$$

with algebraic multiplicities. If both diagonal blocks are Hurwitz, $x \to 0$, $e \to 0$, and $\hat{x} \to 0$ for any initial plant and observer states.

Separation principle. Design a stabilizing K and a convergent observer gain L independently. Stabilizability of (A, B) and detectability of (A, C) are sufficient. Estimation errors still affect plant transients through BKe; separation of poles does not mean identical performance to perfect state feedback.

7 A complete output-feedback controller

7.1 Combine the gains for the second-order example

Use the gain $K = [6, 6]$ from Week 9, which places controller poles at $-2, -4$, and the observer gain $L = [15, 48]^T$, which places observer poles at $-5, -10$. The controller uses only y and its internal state:

$$\boxed{\begin{aligned}&\dot{\hat{x}} = (A - BK - LC)\hat{x} + Ly,\\ &u = -K\hat{x}.\end{aligned}}$$

For the example,

$$\dot{\hat{x}} = \begin{bmatrix}-15 & 1\\ -56 & -6\end{bmatrix}\hat{x} + \begin{bmatrix}15\\ 48\end{bmatrix}y, \qquad u = -6\hat{x}_1 - 6\hat{x}_2.$$

The controller is itself a state-space system with input y and output u. Its internal matrix alone does not determine the poles of the plant–controller interconnection.

7.2 Two equivalent descriptions of the closed loop

In the physical state and estimate coordinates,

$$\begin{bmatrix}\dot{x}\\ \dot{\hat{x}}\end{bmatrix} = \underbrace{\begin{bmatrix}A & -BK\\ LC & A - BK - LC\end{bmatrix}}_{F}\begin{bmatrix}x\\ \hat{x}\end{bmatrix}.$$

Changing coordinates using

$$\begin{bmatrix}x\\ e\end{bmatrix} = \underbrace{\begin{bmatrix}I & 0\\ I & -I\end{bmatrix}}_{T}\begin{bmatrix}x\\ \hat{x}\end{bmatrix}, \qquad T^{-1} = T,$$

gives the triangular matrix TFT^{-1} from the previous page. Thus

$$\det(sI - F) = (s + 2)(s + 4)(s + 5)(s + 10).$$

This identity can be checked numerically without expanding a fourth-degree determinant by hand.

7.3 Initial estimates and implementation

If no prior estimate is available, $\hat{x}(0) = 0$ is a possible choice. The observer converges under the stated assumptions, but a large initial error can produce a large transient. If a position measurement is available, initializing $\hat{x}_1(0)$ from it can reduce the initial discrepancy; the velocity estimate still needs to converge.

At each instant, the controller computes u from $\hat{x}$, the plant evolves using that input, and the observer evolves using the same u together with y. A simulation must evolve both x and $\hat{x}$; feeding the true x to the controller would test a different design.

The separation principle will also apply when K is chosen by LQR. The next design question is how to balance state error and control effort when selecting that gain.

8 Observer speed, transient error, and measurement noise

8.1 Faster poles have a cost

Observer poles are often chosen faster than controller poles, but there is no universal speed ratio. Large gains can increase noise sensitivity and transient peaks even when the error poles are stable.

Let the plant experience an unmodeled disturbance d and measurement noise ν:

$$\dot{x} = Ax + Bu + Ed, \qquad y = Cx + \nu.$$

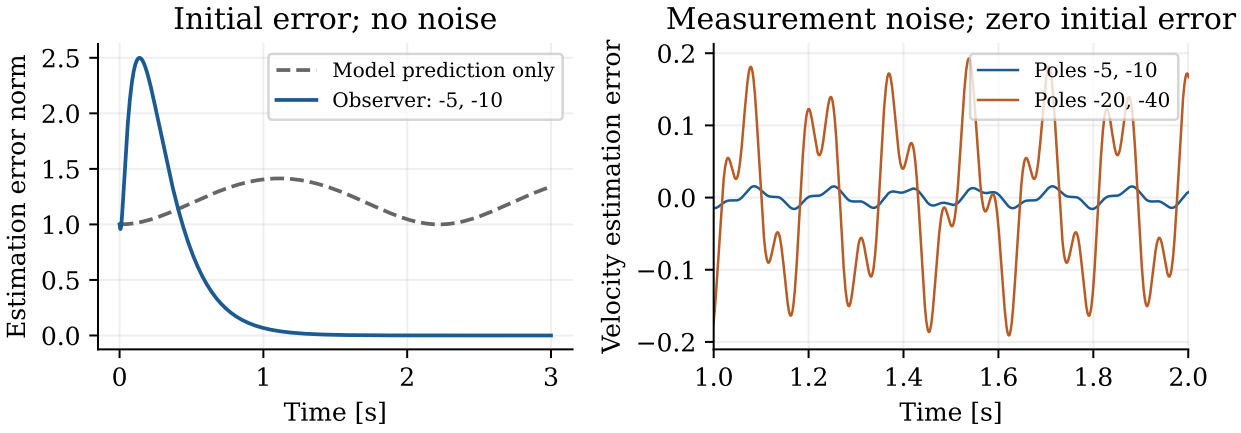
The same observer now gives

$$\boxed{\dot{e} = (A - LC)e + Ed - L\nu.}$$

Stable error dynamics remove initial error, but persistent disturbances or noise generally sustain estimation error. Correction introduces measurement noise through $-L\nu$.

8.2 Compare two observer designs

For the second-order example, poles $-5, -10$ give $L = [15, 48]^T$, while poles $-20, -40$ give $L = [60, 798]^T$. The second design has substantially larger gains, especially in the velocity correction.



Left: exact model, no noise, $e(0) = [1, 0]^T$. The model predictor retains an oscillating error; the observer converges after an initial peak. The norm uses unscaled example coordinates.

Right: $e(0) = 0$ and position noise $\nu(t) = 0.01[\sin(40t) + \sin(110t)]$. The same noise produces larger velocity error in the faster observer. This comparison depends on the noise spectrum; the plot shows the response after the initial transient.

8.3 What to check in a design

Inspect state trajectories, estimation errors, and control effort for plausible initial errors. Repeat with measurement noise, model mismatch, and actuator limits. With saturation, provide the observer with the applied input; the ideal linear separation result does not by itself establish stability of the saturated system.

Kalman filtering will later provide a systematic way to choose correction gains from a stochastic model of uncertainty and sensor noise.

9 Python verification and simulation

9.1 Design the gains and simulate the interconnection

This code uses NumPy, SciPy, and Matplotlib. It reproduces the second-order example with $x(0) = [1, 0]^T$, $\hat{x}(0) = 0$, and noiseless measurements.

```
import numpy as np
from scipy.signal import place_poles, StateSpace, lsim
import matplotlib.pyplot as plt

A = np.array([[0., 1.], [-2., 0.]])
B = np.array([[0.], [1.]])
C = np.array([[1., 0.]])
assert np.linalg.matrix_rank(np.vstack([C, C @ A])) == 2
K = place_poles(A, B, [-2., -4.]).gain_matrix
L = place_poles(A.T, C.T, [-5., -10.]).gain_matrix.T
F = np.block([[A, -B @ K],
              [L @ C, A - B @ K - L @ C]])
print("K:", K, "L:", L.ravel())  # [6, 6] and [15, 48]
print("closed-loop poles:", np.sort(np.linalg.eigvals(F)))
# Expected: -10, -5, -4, -2

t = np.linspace(0., 5., 5001)
nu = np.zeros_like(t)
# For a noise experiment, replace the preceding line with:
# nu = 0.01 * (np.sin(40*t) + np.sin(110*t))
Gnu = np.vstack([np.zeros((2, 1)), L])
sys = StateSpace(F, Gnu, np.eye(4), np.zeros((4, 1)))
_, Z, _ = lsim(sys, U=nu, T=t, X0=[1., 0., 0., 0.])
X, Xhat = Z[:, :2], Z[:, 2:]
E = X - Xhat
U = -(Xhat @ K.T).ravel()
fig, ax = plt.subplots(3, 1, sharex=True, figsize=(6, 6))
ax[0].plot(t, X[:, 0], label="position")
ax[0].plot(t, Xhat[:, 0], "--", label="estimated position")
ax[0].set_ylabel("position"); ax[0].legend()
ax[1].plot(t, E[:, 0], label="position error")
ax[1].plot(t, E[:, 1], label="velocity error")
ax[1].set_ylabel("error"); ax[1].legend()
ax[2].plot(t, U); ax[2].set_ylabel("input u")
ax[2].set_xlabel("time [s]")
fig.tight_layout(); plt.show()
```

9.2 Suggested comparisons

First verify the two gains and all four closed-loop poles. Then compare a correct initial estimate with $\hat{x}(0) = 0$. Finally add the specified noise and repeat with observer poles $-20, -40$, retaining the same K. Rebuild F and G_ν after changing L, and compare estimation error and control effort.

10 Practice and brief answers

10.1 Practice

1. For the example on page 5, place observer poles at $-3, -6$. Find L and write the two observer equations.
2. If A is Hurwitz, does the model predictor $\dot{\hat{x}} = A\hat{x} + Bu$ need output correction to remove an initial error in the exact, noiseless model? What limitations remain?
3. Let $A = \mathrm{diag}(1, -2)$ and $C = [1, 0]$. Is the pair observable? Detectable? Give a convergent observer gain. Repeat the detectability question with $C = [0, 1]$.
4. For the worked controller and observer, write the closed-loop matrix in (x, e) coordinates and identify its four eigenvalues.
5. If $e(0) = 0$ in the exact, noiseless model, how does observer-based feedback compare with full-state feedback? What changes when $e(0) \neq 0$?
6. Suppose $y = Cx + Du$ with known D. Write the correct observer residual. Explain why omitting Du changes the error dynamics.
7. Why does multiplying every observer pole's magnitude by four not guarantee a better practical controller?

10.2 Brief answers

1. $(s + 3)(s + 6) = s^2 + 9s + 18$, so $L = [9, 16]^T$. The equations are $\dot{\hat{x}}_1 = \hat{x}_2 + 9(y - \hat{x}_1)$ and $\dot{\hat{x}}_2 = -2\hat{x}_1 + u + 16(y - \hat{x}_1)$.
2. No: $e(t) = e^{At}e(0) \to 0$ already. Its convergence rate is fixed by A, and disturbances or model errors can sustain discrepancies. Measurement correction permits a different error dynamics when the pair allows it.
3. It is unobservable but detectable, since the unseen pole is -2. Take $L = [4, 0]^T$, giving error poles $-3, -2$. With $C = [0, 1]$, the unseen pole is $+1$, so the pair is not detectable.
4. The matrix is
$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -8 & -6 & 6 & 6 \\ 0 & 0 & -15 & 1 \\ 0 & 0 & -50 & 0 \end{bmatrix},$$
with eigenvalues $-2, -4, -5, -10$.
5. Zero initial error remains zero, so the same initial plant state yields the full-state-feedback trajectory. A nonzero error drives the plant through BKe, changing the transient while retaining stability when both diagonal blocks are Hurwitz.
6. Use $y - C\hat{x} - Du$. Omitting Du introduces the extra term $-LDu$ in $\dot{e}$, so it is no longer the intended autonomous error equation.
7. Faster nominal decay can require much larger gains, increase noise sensitivity and transient peaks, and increase control effort. Performance depends on sensor noise, uncertainty, initial estimates, and actuator limits as well as poles.